Dungeon Hunter: MVP PRD

Oct 23, 2026 · @diaz

Overview

The MVP is one complete, replayable run through the Hollow Vault: three sectors, two sector bosses, and the Core Warden, playable in a desktop browser in 20 to 30 minutes.

Dungeon Hunter is a top-down pixel roguelite shooter in the style of Soul Knight, rebuilt for mouse and keyboard. Each run is a new Runner deployment into a facility that reshuffles itself; death ends the run, Core Shards carry over.

Pillars

1. **Guns feel great.** Every shot has recoil, sound, shake, and hit feedback. If shooting a wall isn't fun, nothing else matters.
2. **Short runs, endless replay.** Procedural sectors, random weapons and Vault Mods. A run takes under 30 minutes.
3. **Story told by the Vault.** Light narrative through radio lines, terminal logs, and environments. Never blocks gameplay.

Target player: desktop players who enjoy Soul Knight, Enter the Gungeon, or Nuclear Throne and want a quick session in the browser with no install.

MVP is done when: a new player can sign up, pick a Runner, clear all three sectors, beat the Core Warden, see the ending, earn Core Shards, unlock something, and appear on the leaderboard.

Open question: "Dungeon Hunter" is already a well-known game franchise (Gameloft). Fine as a working title, but pick a different public name before launch. *Hollow Vault* or *Vault Runner* fit the story.

Story and world

The story is your premise, kept intact, with three additions that make it work as a roguelite: a voice on the radio, a villain inside the system, and a reason dying is part of the fiction.

World

- **Veyra:** a dense city whose lights run on power drawn from below.
- **The Hollow Vault:** an old underground energy-storage facility under Veyra, built in three levels: Storage Wing, Conduit Works, Core Approach.

- **The Core:** the energy source at the bottom. It is corrupting, and the corruption (called **the Bleed**, a magenta glow that warps machines) is climbing toward the surface.
- **VIGIL:** the Vault's security system. Its malfunction woke the guardian machines and sealed the paths to the Core.

Characters

Name	Role	How the player meets them
The Runner	Player character, a specialist sent by Veyra's Grid Authority to shut the Core down	Playable; each run is a new deployment ("Runner Unit 047")
Ops	Grid Authority handler on the radio	Short text lines at sector starts, boss fights, death, ending
VIGIL	Malfunctioning security system	Terminal logs and warnings on screen ("INTRUDER. SECTOR SEALED.")
Core Warden	Final guardian, fused with the corrupted Core	Final boss in the Core Chamber

Why death fits the story: Ops deploys a new Runner after each lost signal. The results screen reads "Runner signal lost. Deploying next unit." Core Shards are what the Grid Authority salvaged from the last attempt.

The small twist (Sector 3 logs): VIGIL didn't just glitch. The Bleed reached VIGIL first, and it now protects the Core because it reads the corruption as the Core's "normal" state. The Core Warden isn't defending the city, it's defending the infection. Shutting the Core down also frees VIGIL.

Story beats in a run

Moment	What the player sees
First launch	4 text cards over pixel art, following your Pixar beats (Veyra, the Vault, the malfunction, the Runner)
Sector 1 entry	Ops: mission briefing. VIGIL: "INTRUDER DETECTED."
Gatekeeper defeated	Ops: path to the Conduit Works open
Sector 2 entry	Ops notices the Bleed spreading in the machines
Conduit Hydra defeated	VIGIL message glitches, first hint of the twist

Moment	What the player sees
Sector 3 entry	Environment turns magenta; Ops loses signal clarity
Core Chamber	VIGIL speaks directly before the Warden wakes
Victory	Core shuts down, VIGIL's last clean message, Veyra's lights steady. Ending card + credits
Death	Ops: signal lost. Results screen

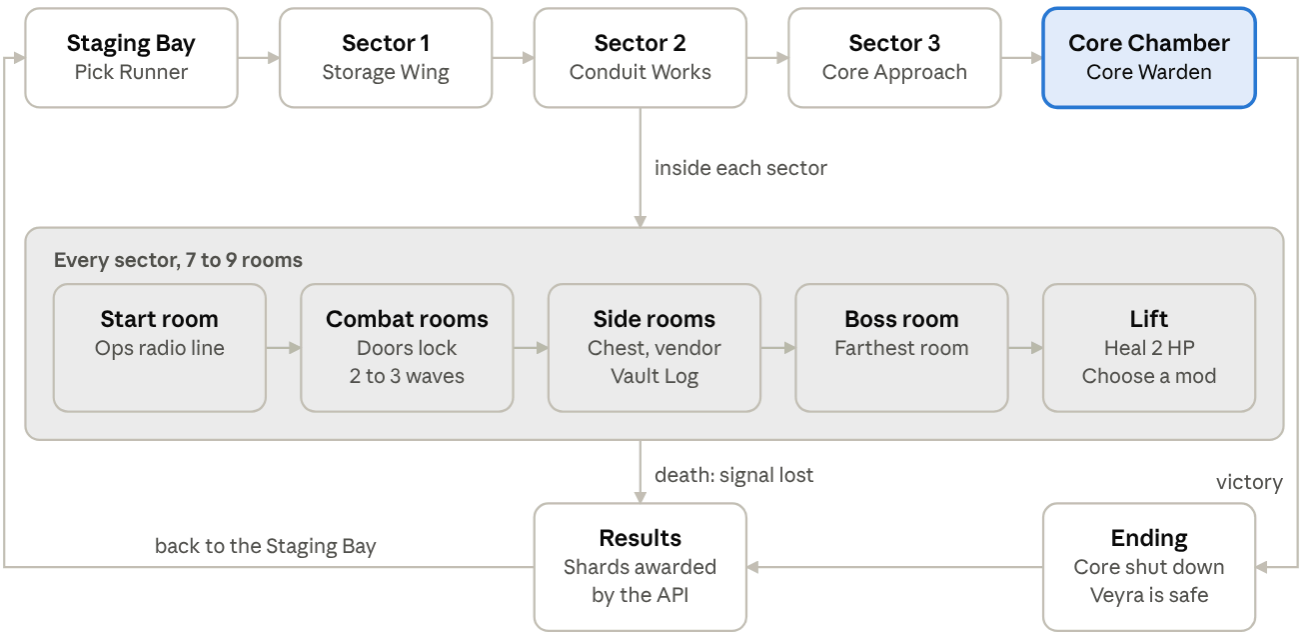
Vault Logs: 9 collectible terminals (3 per sector, one may spawn per sector per run). Each is 2 to 4 lines of lore. Collected logs are saved to the profile and readable in the Staging Bay. This is the main way the twist is told.

Rule: story text never pauses combat. Radio lines appear as a small box at the bottom of the screen and fade on their own.

Core gameplay loop

A run goes down through three sectors to the Core Chamber; it ends in the Core Warden's defeat or the Runner's death, and either way the player returns to the Staging Bay with Core Shards.

Three sectors, one Warden, then back to the Staging Bay



run loop · 3 sectors, 1 final boss

Every sector repeats the same rhythm: clear combat rooms, grab side-room loot, beat the boss, take the Lift down. Death anywhere ends the run.

Target pacing: Sector 1 about 8 minutes, Sector 2 about 9, Sector 3 plus the Warden about 10. A death in Sector 1 should still feel worth it because Core Shards are paid per room cleared.

Runners, controls, combat feel

The MVP ships two Runners with different skills; both use the same three resources Soul Knight players already understand: HP, Shield, Energy.

Resources

- **HP:** lost only after Shield is empty. Restored by health capsules and Lift rooms.
- **Shield:** absorbs hits, regenerates 1 point per second after 3 seconds without damage.
- **Energy:** ammo for every weapon except the starter pistol and melee. Dropped as orbs by enemies and crates.

Runners (all numbers are starting values, tuned in playtest)

Runner	HP	Shield	Energy	Skill (right click)	Unlock
Vex, Breacher	6	4	180	**Overcharge:** double fire rate, no energy cost for 5 s. Cooldown 20 s	Free
Kit, Tinker	5	5	200	**Patch Drone:** deploys a turret that auto-fires for 8 s. Cooldown 22 s	300 Core Shards

Controls

Action	Keyboard + mouse	Gamepad
Move	WASD	Left stick
Aim	Mouse	Right stick
Shoot	Left click (hold)	Right trigger
Skill	Right click or Space	Left trigger
Dash (short i-frames)	Shift	A / Cross
Swap weapon	Q or mouse wheel	Y / Triangle
Interact, pick up	E	X / Square
Pause	Esc	Start

Carry 2 weapons. Picking up a third drops the current one on the floor.

Aim assist: mouse aim by default, with soft assist (bullets bend up to 8 degrees toward an enemy near the cursor). Settings has "Full auto-aim" for the mobile-style experience.

Combat feel checklist (must ship in M1, not polish)

- Screen shake per weapon (small for SMG, big for launcher)
- Enemy hit flash (white, 50 ms) + knockback
- Hit-stop on boss hits and kills (30 ms)
- Muzzle flash, shell casings, impact sparks
- Distinct sound per weapon type, pitch-randomized
- Custom crosshair cursor
- Damage numbers off by default (toggle in settings)

Weapons and Vault Mods

The MVP has 12 weapons and 10 Vault Mods, all defined in JSON in `packages/shared/src/data` so new ones need no new code.

Weapons

Weapon	Type	Rarity	Energy per shot	Behaviour
Rust Pistol	Pistol	Starter	0	Reliable, infinite ammo
Shock Baton	Melee	Common	0	Short swing, stuns 0.5 s

Weapon	Type	Rarity	Energy per shot	Behaviour
Rivet SMG	Automatic	Common	1	High fire rate, wide spread
Breach Shotgun	Shotgun	Common	3	5 pellets, strong knockback
Line Rifle	Rifle	Common	2	Pierces 1 enemy
Arc Carbine	Burst	Rare	2	3-round burst
Scatter Coil	Shotgun	Rare	4	Pellets bounce off walls once
Pulse Lance	Beam	Rare	5 per second	Continuous laser, ramps damage
Grid Cutter	Melee	Rare	0	Swing deletes enemy bullets
Fuse Launcher	Launcher	Rare	6	Explosive shell, area damage, self-safe
Static Repeater	Automatic	Epic	1	Shots chain lightning to 2 nearby enemies
Null Cannon	Charge	Epic	10	Hold to charge, huge piercing orb

Starter pool: the 5 Common weapons + Rust Pistol. Rare and Epic weapons enter the drop pool once unlocked in the Armory.

Vault Mods are passive upgrades. After each sector boss and from gold chests, the player picks 1 of 3.

Mod	Effect
Cooling Fins	+20% fire rate
Spare Cell	+1 max Shield
Battery Pack	+60 max Energy
Overclock Core	+25% damage, -1 max HP
Ricochet Chip	Bullets bounce off walls once
Scrap Magnet	Pickup radius doubled
Dash Coil	+1 dash charge
Field Patch	Heal 1 HP when a sector boss dies
Thorn Plating	Enemies that touch you take damage

Mod	Effect
Lucky Seal	Higher rarity chance on chests

Rule for adding content later: a new weapon or mod is a JSON entry plus, at most, one new projectile behaviour.

Guardians and bosses

The MVP has 7 guardian machine types, 2 sector bosses, and the Core Warden; Sector 3 reuses earlier guardians as Bleed-corrupted variants instead of needing all-new enemies.

Guardian machines

Guardian	Sector	Behaviour	Teaches the player
Sentry Drone	1+	Hovers, fires a single slow bolt	Dodge and shoot back
Crawler	1+	Rushes at the player, explodes on contact	Keep moving, prioritize threats
Turret Node	1+	Stationary, fires 3-way spread	Use cover
Shield Bot	2+	Frontal shield blocks bullets	Flank or use bouncing shots
Arc Spider	2+	Fast, leaves an electric trail	Control space
Mortar Unit	2+	Lobs shells at your position, telegraphed circle	Don't stand still
Bleed Wraith	3	Teleports, fires a ring of bullets	Read patterns

Bleed variants (Sector 3): magenta tint, +40% HP, faster bullets, and on death they leave a short-lived damaging puddle. One shader + stat multiplier, no new art.

Elites: any guardian can spawn as an elite (bigger, 2x HP, drops a guaranteed energy orb). 1 per room maximum.

Sector bosses

- **Gatekeeper (Sector 1):** a heavy security tank guarding the Storage Wing lift. Pattern A: charges in a straight line and stuns itself on walls (damage window). Pattern B: stationary bullet ring. Below 50% HP: charges twice in a row.

- **Conduit Hydra (Sector 2):** a core with 3 cable heads. Each head fires a different pattern (aimed shots, spread, beam sweep). Destroying a head removes its pattern; the core becomes vulnerable after all 3.

Core Warden (final boss)

The Warden is fused with the Core and fights in three phases, each longer and harder:

1. **Shielded:** 2 rotating shield arms block shots from the front. Fires spiral patterns. Players must circle around to hit the body.
2. **Lockdown:** VIGIL seals the arena edges (shrinking floor of electrified tiles) and the Warden summons 2 waves of Bleed guardians.
3. **Core Exposed:** the arms break, the Core is visible. Dense bullet-hell spirals, faster as HP drops. At 15% HP, a 3-second scream telegraph, then a final enraged pattern.

Winning triggers the shutdown sequence and the ending. Target: first-time victory rate around 10 to 20% in playtests; experienced players should win most runs.

Dungeon structure

Each sector is generated from hand-made Tiled room templates placed on a 5x5 grid with a seeded RNG, so every run differs but every room is designed.

Sectors

Sector	Name	Look	Guardians	Boss
1	Storage Wing	Rust and amber, crates, cargo rails	Drone, Crawler, Turret	Gatekeeper
2	Conduit Works	Teal pipes, steam, cable bundles	All of Sector 1 + Shield Bot, Arc Spider, Mortar	Conduit Hydra
3	Core Approach	Magenta Bleed glow, warped metal	Bleed variants + Bleed Wraith	Core Warden (Core Chamber)

Room types per sector (7 to 9 rooms)

Room	Count	Purpose
Start	1	Safe, radio line from Ops
Combat	4 to 6	Doors lock, 2 to 3 waves, unlock on clear, drop Scrap and orbs

Room	Count	Purpose
Chest	1	Weapon chest (sometimes gold: Vault Mod choice)
Vendor terminal	1 (Sectors 2 and 3)	Spend Scrap on weapons, health, energy
Log terminal	0 or 1	Vault Log collectible
Boss	1	Farthest from Start, dead end
Lift	after boss	Heal 2 HP, choose 1 of 3 Vault Mods, go down

Generation rules

- Start at grid center; random walk places rooms; corridors are 3 tiles wide.
- Boss room = farthest room by path length, minimum 4 rooms from Start.
- Chest and Vendor never adjacent to Start.
- Validate every layout with a reachability check; regenerate on failure (unit-tested).
- Difficulty budget per room = sector base + rooms cleared. Waves spend the budget on guardians (Drone = 1, Crawler = 1, Shield Bot = 3, and so on).
- Seed shown on the results screen so players can share runs.

Content target: 15 combat room templates (5 per sector theme), 1 template each for the special rooms, 3 boss arenas. Rooms are 15x11 tiles of 16 px, fitting the 480x270 view with a small margin.

Economy and meta progression

Two currencies: Scrap lives and dies inside a run, Core Shards persist and are the only thing the server awards.

Currency	Earned from	Spent on	Persists
Scrap	Guardians, crates, rooms cleared	Vendor terminals (weapons, health, energy)	No, reset each run
Core Shards	End-of-run payout, calculated by the API	Armory unlocks in the Staging Bay	Yes, stored in Supabase

Core Shard payout (server-side, starting formula)

- 2 per room cleared
- 25 per sector boss
- 100 for defeating the Core Warden

- Capped by run duration and rooms cleared (see anti-cheat rules in the architecture section)

Armory unlocks (starting prices, tuned in playtest)

Unlock	Cost
Kit, Tinker (Runner)	300
Each Rare weapon into the drop pool	80
Each Epic weapon into the drop pool	150
Extra Vault Mods into the pool (Thorn Plating, Lucky Seal, Dash Coil)	60 each

Also persistent, but free: collected Vault Logs, best time, total runs, total victories.

No real-money purchases, no energy timers, no gacha in the MVP.

Screens and UX flow

React owns every screen outside a run; Phaser owns the run itself, including the HUD, so the two never render at the same time.

Screen	Built with	Contents
Title	React	Logo, Play, Continue as guest, Log in, Settings
Intro cards	React	4 story cards on first launch, skippable
Login / Register	React	Email + password through the API; guest mode keeps progress locally
Staging Bay (lobby)	React	Runner select, Armory, Vault Logs, Leaderboard, Start run
Armory	React	Unlock Runners, weapons, mods with Core Shards
Leaderboard	React	Weekly and all-time; fastest victories, deepest runs
Settings	React	Volume, aim mode, screen shake, damage numbers, keybind reference
In run	Phaser	Dungeon scene + HUD scene (HP, Shield, Energy, weapons, skill cooldown, minimap, Scrap)
Pause	Phaser	Resume, settings, abandon run

Screen	Built with	Contents
Results	React	Victory or signal lost, stats, Core Shards earned, seed, new Vault Logs
Ending	React	Shutdown sequence cards + credits, shown on first victory

Guest vs account: guests play fully with progress in localStorage. Registering offers a one-time import of local progress (capped values, validated by the API). Leaderboard requires an account.

Minimap: top-right, reveals rooms as they're seen, icons for chest, vendor, boss. Toggle full map with Tab.

Art and audio direction

Cold industrial pixel art with one loud colour for danger: the Bleed's magenta grows from a hint in Sector 1 to the whole screen in Sector 3, so the story is visible without reading.

Visual specs

- Internal resolution 480x270, scaled with nearest-neighbour; `pixelArt: true`
- Tiles 16x16; Runners 16x24; guardians 16x16 to 32x32; bosses 64x64 to 96x96
- Palette: 32 colours max. Base greys and steel blues, sector accent (amber, teal), Bleed magenta reserved for corruption and danger
- Enemy bullets always bright and outlined; player bullets warmer and smaller, so threats read instantly
- Pixel font for UI (both React screens and HUD use the same font)

Asset plan

- Prototype with free packs (for example 0x72's Dungeon Tileset II on itch.io; check each pack's licence before shipping)
- Replace with original art from M5 onward, Runners and bosses first since they define the game's identity
- Do not use any Soul Knight sprites, names, or recognisable character designs

Audio

- Music: synth/chiptune with industrial percussion. 1 lobby track, 1 track per sector, 1 boss track, 1 Core Warden track (6 total)
- SFX priority: weapon shots, hits, enemy deaths, door lock/unlock, pickups, VIGIL alarm
- Every SFX pitch-randomized ±5% so repeated shots don't sound robotic

Technical architecture

The browser never talks to Supabase: React and Phaser call the Hono API, Hono holds the secret key and every rule about currency, unlocks, and scores. Full setup is in SETUP.md.

Layer	Tech	Responsibility
Game	Phaser + TypeScript	The run: movement, combat, AI, generation, HUD. Runs fully offline
UI	React + Vite + Tailwind	All screens outside a run
Client state	Zustand	Settings, selected Runner, menu state
Server state	TanStack Query + Hono RPC client	Profile, unlocks, leaderboard, saves
API	Hono on Cloudflare Workers + Zod	Auth, validation, Core Shard payouts, purchases
Database + auth	Supabase (Postgres + Auth)	RLS on, no public policies; reachable only from the API
Shared	`packages/shared`	Types, Zod schemas, weapons/enemies/mods JSON
Tests	Vitest	Generator, damage math, run validation, API routes

When the game talks to the API: app start (load profile), run end (submit result), Armory purchase, register/login. Nothing during combat.

Data model

- `profiles` : id, username, core_shards, created_at
- `unlocks` : profile_id, item_type, item_id
- `vault_logs` : profile_id, log_id, found_at
- `saves` : profile_id, slot, data (jsonb with a version field), updated_at (post-MVP mid-run saves; MVP stores settings only)
- `runs` : profile_id, runner_id, seed, sector_reached, rooms_cleared, bosses_killed, duration_sec, kills, victory, shards_awarded, created_at

API endpoints (MVP)

Method	Path	Purpose
POST	/api/auth/register	Create account, set httpOnly session cookie
POST	/api/auth/login	Log in, set cookie
POST	/api/auth/logout	Clear cookie
GET	/api/me	Profile, Core Shards, unlocks, Vault Logs
POST	/api/me/import	One-time guest progress import (capped)
POST	/api/runs	Submit run result, server computes shards awarded
POST	/api/armory/purchase	Buy an unlock; server checks price and balance
GET	/api/leaderboard	Weekly or all-time rankings

Anti-cheat rules in `POST /api/runs`

- Client sends what happened, never how many shards it earned. The server calculates the payout.
- Reject if rooms_cleared exceeds the maximum for sector_reached, or bosses_killed exceeds sector_reached.
- Reject if duration_sec is below a minimum per room cleared (start at 8 s per room) or victory comes under 6 minutes.
- Rate limit: 1 run submission per 60 s per account.
- Leaderboard shows only runs that passed validation.

This stops casual devtools cheating, not determined hackers. Good enough for an MVP.

MVP scope and milestones

Eight milestones, roughly 16 weeks for one developer working part-time; gameplay is fully playable offline by M4, and the backend only starts at M6.

In scope: everything listed in this document: 2 Runners, 12 weapons, 10 Vault Mods, 7 guardians, 2 bosses + Core Warden, 3 sectors, story beats and 9 Vault Logs, accounts, Armory, leaderboard, desktop browser with keyboard/mouse and basic gamepad.

Out of scope for MVP: multiplayer/co-op, mobile and touch controls, more than 2 Runners, mid-run save and resume, pets, cosmetics, achievements, voice acting, localization, keybind remapping, real-money purchases.

Milestone	Est.	Delivers	Done when
M0 Setup	1 wk	Monorepo per SETUP.md, CI running typecheck, lint, tests	`pnpm dev` runs web + api; a test passes in CI
M1 Core combat	2 wk	Movement, dash, mouse aim + assist, 3 weapons, pooled bullets, 2 guardians, full juice checklist	Shooting a Crawler feels good to 3 testers; 300 bullets at 60 fps
M2 Rooms	2 wk	Tiled templates, door lock/unlock, waves, seeded generator, minimap	100 generated seeds all pass reachability tests
M3 Sector 1 vertical slice	2 wk	Sector 1 content, Gatekeeper, chests, Vault Mods, HUD, Lift	A full Sector 1 run start to boss, 8 to 10 minutes
M4 Full run	3 wk	Sectors 2 and 3, remaining guardians, Bleed variants, Hydra, Core Warden, all 12 weapons	A complete offline run is winnable
M5 Story + menus	2 wk	Intro cards, radio lines, Vault Logs, ending, React screens, Kit, original Runner and boss art	First-time player understands the goal without explanation
M6 Backend	2 wk	Hono auth, runs, payouts, Armory, leaderboard, guest import, RLS lockdown	Shards survive a browser wipe; a forged run submission is rejected
M7 Polish + launch	2 wk	Balance pass, audio, playtests (10+ people), deploy to Cloudflare + Supabase	Public URL live; no crash in 20 consecutive runs

Cut order if time runs short: Kit (ship with 1 Runner), Vendor terminals, Vault Logs (keep radio lines), gamepad. Never cut the combat feel checklist or the Core Warden.

Risks and success metrics

The biggest risk is scope, not tech: the stack is solid, but content (rooms, art, boss patterns) is what eats solo projects.

Risk	Impact	Mitigation
Scope creep	MVP never ships	Strict out-of-scope list; cut order above; no new features before M7
Combat feels flat	Nobody replays	Juice checklist is part of M1, playtest gun feel before building rooms
Art bottleneck	Delays M5	Free packs for prototyping, original art only for Runners and bosses first
Performance with many bullets	Stutter on laptops	Object pooling from day one, profile on a mid-range laptop every milestone
Leaderboard cheating	Unfair rankings	Server-side payout and validation rules; accept that MVP stops only casual cheats
Name / IP conflict	Forced rename after launch	Rename before public launch; no Soul Knight assets or names

Success metrics for MVP launch

Metric	Target
Frame rate on a mid-range laptop, 300 bullets on screen	60 fps
First load time	Under 5 s
Average run length	20 to 30 min
First-time Core Warden victory rate (playtest)	10 to 20%
Playtesters who start a second run without being asked	70%+
Crashes in 20 consecutive runs	0